Case Study: *C. trachomatis* — Diagnostic Target Selection and In-Silico Oligonucleotide Screening

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Executive summary

This project computationally verified a panel of diagnostic candidate targets for a multiplex qPCR/TaqMan assay for *Chlamydia trachomatis*. Four chromosomal loci (mutL, aroB, PSD2, and group_306) were retained as candidate targets, represented by seven independently screened oligo sets, alongside two candidate windows on the organism's multicopy diagnostic plasmid. The proposed duplex assay combines one chromosomal locus with the plasmid target, providing an independent genomic marker alongside the multicopy target, which has the potential to improve analytical sensitivity.

Candidate sets were evaluated using a common acceptance framework covering calculated primer/probe Tm and Tm gap, standardized BLAST specificity against the human genome and nucleotide databases, pair-level off-target analysis, and three-way structural assessment of all primer–probe interactions. All nine retained sets passed the defined in-silico acceptance criteria. The work is computational: amplification efficiency, analytical sensitivity, specificity in real samples and multiplex performance remain to be established experimentally.

A central feature of the workflow is independent verification of intermediate results. Recalculation of inherited Tm values, examination of genome and plasmid alignments, and independent assessment of candidate-ranking criteria identified several issues that would otherwise have propagated into the final design. The resulting workflow therefore combines target discovery with explicit QC gates and traceable decision criteria.

Background and objectives

Chlamydia trachomatis is an important target for nucleic-acid amplification testing, where assay performance depends on selecting sequences that are both conserved across the target population and sufficiently differentiated from related organisms and human DNA. A duplex design combining a conserved chromosomal locus with the organism's multicopy diagnostic plasmid provides two independent target classes: a stable genomic marker and a higher-copy target with the potential to improve analytical sensitivity.

The objective of this project was computational: to identify conserved candidate regions and rigorously evaluate primer/probe sets for inclusivity, exclusivity, thermodynamic suitability and structural compatibility before experimental testing. Two structurally different target classes were pursued: four chromosomal genes (mutL, aroB, PSD2, group_306) and the species' multicopy cryptic plasmid, each needing its own genome curation, consensus-building and design pipeline.

Methodology: from raw genomes to a candidate panel

The 125 RefSeq genomes used throughout this pipeline are the result of a provenance audit: the project's original genome inventory also included 149 GenBank-only complete genomes, which a BioProject-level audit traced to a single study of deliberately engineered laboratory strains — transposon-mutant and antibiotic-resistance-tagged constructs — rather than natural clinical isolates, and excluded outright. The 125 RefSeq genomes are the clean baseline this pipeline builds from. Even so, none of the 125 were accepted on RefSeq status alone — each was independently re-verified and tiered into primary, quality-review, or excluded sets, as the Genome curation and Provenance audit rows below show.

Stage	Genome count	What it means
Provenance audit (bin/02)	125	Full RefSeq set, tiered: 97 primary / 10 quality-review / 18 excluded
Prokka annotation (bin/05)	107	Primary + quality-review, annotated
Panaroo pangenome (bin/06)	97	Primary tier only (904 gene clusters, 874 core); quality-review genomes are not pangenome inputs
Consensus-building / target sequence (bin/08–09)	97	The same 97 primary-tier genomes build the consensus each candidate target's sequence is drawn from

107 is the count that enters annotation; 97 is the count that drives everything after it — the pangenome, each target's consensus sequence, and the inclusivity claim itself. Quality-review genomes are annotated so they can still be examined later, but they are kept out of the pangenome and the consensus rather than allowed to shape them. The 18 excluded genomes never enter any of this past the provenance audit. On the plasmid target, the quality-review genomes that carry a plasmid were aligned alongside the primary set and checked against the consensus: all sit at 99.95–100% identity to it. On the chromosomal panel, the ten quality-review genomes were tested directly against the

finished oligo sets (bin/18, BLAST of all 21 oligos with the 97 primary genomes as a control): 69 of 70 set-by-genome cases bound exactly, one (a probe with a single internal mismatch in a backup set) bound within tolerance, and none was at risk or failed.

The ten quality-review genomes split into three distinct flags, not one generic concern. Five were flagged on raw assembly metrics measured against the panel: two carry a CheckM contamination score of 7.15% and a chromosome length roughly 41,100 nt longer than the panel median of 1,042,743 bp; one is 19,111 nt longer with no contamination flag; and two others are 16,904 nt and 9,511 nt shorter, with CheckM completeness scores of 91.61% and 93.78% respectively (completeness below 100% plus a shorter-than-median length both point the same way: a genuinely incomplete assembly, not a sequencing artifact). Four more — all from a single BioProject whose own title records "Laboratory adaptation of Chlamydia trachomatis" — are genuine clinical isolates that had been serially passaged in culture, which can introduce drift even though the original infection was natural. The tenth carries no flagged assembly metric at all; its BioProject metadata names a different organism entirely (Bordetella pertussis) against a C. trachomatis-labeled strain — a submission-metadata inconsistency flagged for exactly that reason, not a sequence-quality one.

The chromosomal pipeline ran as a sequence of purpose-built scripts, each producing the input the next stage needed:

Stage	Script	What it produced
Genome curation	bin/01	125 RefSeq *C. trachomatis* genomes downloaded and QC'd
Provenance audit	bin/02	Tiering: 97 primary, 10 quality-review, 18 excluded
Species + redundancy check	bin/03	fastANI: all pairs ⩾98.90% identity, supporting species-level consistency: every genome is consistent with *C. trachomatis*
ompA serovar typing	bin/04	14 serovar groups identified
Annotation	bin/05	Prokka gene calls across 107 genomes
Pangenome	bin/06	Panaroo: 904 gene clusters, 874 core

Stage	Script	What it produced
Exclusivity screening	bin/07	692 of 874 core genes with a gap of at least 150 bp below 85% identity to every genome in the 123-genome exclusivity panel
Candidate region + consensus	bin/08–09	A majority-consensus sequence per candidate region
Primer3 design + QC	run manually against the consensus windows	Oligo sequences, real-computed Tm, structure
BLAST specificity screen	manual web BLAST	Specificity against the human genome and core_nt

The plasmid pipeline (bin/12–17, added later in the project) followed the same shape from a different starting point: 34 of the 125 genomes turned out to already carry an assembled ~7.5kb second contig — the species' cryptic plasmid — so no fresh genome curation was needed, only extraction, orientation correction, consensus-building and design against that real data.

Both pipelines converge on the same acceptance standard, covered next.

Independent verification and redesign

An earlier analysis pass, dated September 24–26, 2026, reported all 44 oligos across the four chromosomal genes (5 forward-primer ranks + 5 reverse-primer ranks + 1 probe per gene) as validated and "ready for wet-lab validation." It scored each oligo on a custom 0–100 exclusivity scale and screened secondary structure with a hand-built hairpin and homodimer detector; 95% of oligos scored ≥95/100, and the write-up's stated next step was a GitHub push followed by wet-lab testing.

Independently recomputing Tm and the Tm gap with primer3-py — the calculation the finished panel actually relies on — found that every one of the four genes failed the project's own ≥5°C Tm-gap floor: PSD2's gap was about 0.2°C (probe 63.8°C against a 66°C floor), mutL's 2.6–3.1°C, group_306's 2.7°C, and aroB's was negative — its probe, at 58.6°C, ran 7–9°C below its own primers rather than above them. A second, independent nearest-neighbor calculation (Biopython's Tm_NN) agreed with primer3-py to within about 1°C, ruling out a single-tool artifact.

The aggregate 0–100 score did not capture this failure mode because Tm gap was not included as an independent acceptance criterion: it rewarded the absence of off-target BLAST hits, which these oligos genuinely had, but carried no term for Tm gap at all, so a set could score 99/100 while failing the predefined Tm-gap requirement for a TaqMan trio. Its

structural check was narrower too — forward–reverse dimers only, never the probe's interactions with either primer (F-Probe, R-Probe), which matter just as much in a three-oligo set.

None of this pointed to a specificity problem: the BLAST findings behind the original scores held up under re-verification. It meant the *design* needed to be redone, not the target selection. Rerunning primer3-py on the exact same, already-verified conserved-window coordinates produced replacement sequences meeting every predefined acceptance criterion for all four genes on the first pass — Tm gaps of 7.54–8.07°C — which is what let the project keep the earlier region-selection work rather than repeat it from scratch.

Standardized oligo acceptance framework

Each gap above turned into a standing rule, logged in the project's own decision record rather than left as a one-off fix:

Requirement	Earlier ad hoc approach	This project's standard
Tm / Tm gap	Read from a settings file's target values, or an uncomputed estimate	primer3-py `calc_tm`, computed on the real candidate sequence; ≥5°C hard floor, ≥7°C preferred
BLAST parameters	Not standardized run to run	Fixed: Expect=1000, word_size=7, filter off, against the human genome and core_nt (excl. *C. trachomatis*'s own taxid)
Off-target logic	Any single-oligo hit counted against the set	A hit is a predicted amplicon only if the forward and reverse primers both bind nearby on the same accession, correctly oriented; the probe then determines whether that product could give signal. A probe plus one primer alone is not a predicted amplicon
Structural check	A hand-written hairpin/dimer detector, forward–reverse pairs only	primer3-py `calc_hairpin` / `calc_homodimer` / `calc_heterodimer` on all three pairs (F-R, F-Probe, R-Probe), severity set by the structure's own melting Tm
Aggregate scoring	One 0–100 score blending unrelated failure modes into a single number	No blended score — Tm gap, BLAST cleanliness and structure are reported and must each pass on their own

The same five checks — real Tm, standardized BLAST, pair-level logic, three-way structure, no blended score — were then applied, unchanged, to the second target class.

Extending the method: the plasmid target

A preliminary plan proposed curating a separate collection of plasmid records. Inspection of the existing genome dataset identified a more direct route: 34 of the 125 RefSeq assemblies already contained an assembled ~7.5kb second contig (7,415–7,510bp across all 34) corresponding to the species' cryptic plasmid. That existing, already-verified starting point replaced the proposed fresh curation entirely.

Aligning the 34 raw contigs with MAFFT produced an alignment roughly twice the length of the input sequences — a strong signal, given how closely length-matched the inputs were, that something other than ordinary divergence was going on. The cause was RefSeq's own submission history: a circular plasmid gets linearized at an arbitrary point,

and different submissions started (and in some cases oriented) it differently. Six anchor probes against a reference genome were used to detect each sequence's rotation and strand, correcting 31 of 34 (3 were flagged as low-confidence and excluded — the same 3 that an earlier, unrelated chromosomal QC pass had already flagged independently). Re-aligning the corrected 31 gave a 7,606bp alignment, in line with the raw sequence length.

A majority-rule consensus from the 27 primary-tier genomes came out 98.3% identical across all of them, with the remaining 1.65% correctly N-masked as real cross-strain variation. That consensus also settled a specific factual claim from the original handover — that the plasmid runs about 50% GC, higher than the ~43% chromosomal figure, which had been used to justify a higher probe-Tm target. The measured consensus GC is 35.4%, *lower* than the chromosome, not higher, so the design used the same Tm settings as the chromosomal panel instead. Primer3 design against the two longest N-free windows of the consensus (4,860–5,277 and 1,315–1,612) then went through the identical D6–D21 standard as every chromosomal set. These 34 plasmid sequences come from the curated genome collection, not from a survey of plasmid diversity, so conservation is shown across the sequences examined only. The plasmid design is a standard consensus design; the nvCT variant, a documented plasmid deletion, is not covered, because its published coordinates could not be independently confirmed (see Scope and limitations).

Verification checkpoints in practice: three issues caught before candidate selection

Beyond the chromosomal redesign above, three more concrete errors were caught during the plasmid work by refusing to treat an intermediate result as correct just because it ran without error:

Issue	What it would have caused, unnoticed	How it was caught	Fix
Circular-genome alignment doubling	A consensus built from a 13,075bp alignment of ~7,500bp sequences — largely spurious gap columns, not real biology	The alignment length itself was implausible for sequences that closely length-matched	Anchor-probe rotation/strand correction before alignment; the corrected re-alignment (7,606bp) confirmed the fix

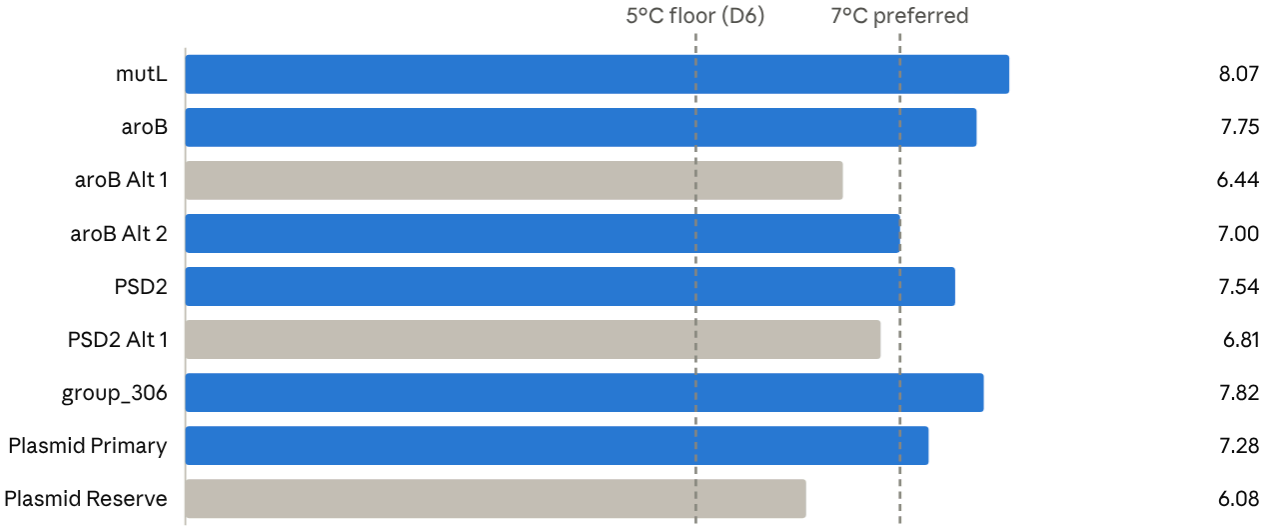
Issue	What it would have caused, unnoticed	How it was caught	Fix
Plasmid candidate ranking rule	A Tm-gap/length-only ranking would have locked in a candidate with a flagged MEDIUM-severity primer hairpin over an equally viable, fully clean alternative — in both design windows	Computing the full three-way dimer check for every surviving candidate, not only the nominal top pick	Re-ranked by structural cleanliness first, then Tm gap and probe length (logged as decision D24)
Plasmid GC% claim	A higher, incorrect probe-Tm target, adopted on the strength of an unverified "~50% GC" claim	Measuring the actual consensus GC directly (35.4%) instead of trusting the figure	Kept the same Tm settings already verified on the chromosomal panel

In each case the error was invisible from the output of the step that produced it — it only showed up once that output was checked against an independent computation or a plausibility test, rather than passed forward as given.

Final results: the in-silico candidate panel

All nine sets clear the 5°C Tm-gap floor

6 of 9 sets also clear the 7°C preferred margin



Tm gap = probe Tm minus the higher primer Tm (°C). Bars start at zero.
Blue: at or above the 7°C preferred margin. Grey: between the 5°C floor and 7°C.

computed with primer3-py · 9 candidate sets

The nine sets break down as seven chromosomal sets across four genes — one for mutL, three independent loci for aroB (primary, Alt 1, Alt 2), two for PSD2 (primary, Alt 1), and one for group_306 — plus two candidate windows on the plasmid (Primary, Reserve).

The panel's nine sets are computationally verified diagnostic candidates, not a single nine-oligo reaction: the proposed duplex assay deploys one chromosomal locus alongside the plasmid target, with the rest held as backups. Every locked set clears the Tm-gap floor, and the two structural/specificity checks below hold for all nine — no plausible off-target amplicon is predicted in the human genome or against non-target *Chlamydia* species for any set, under the defined screening conditions (BLAST parameters fixed project-wide; a predicted amplicon requires both primers).

Target / set	Panel	Structure (three-way dimer check)	BLAST specificity
mutL	Chromosomal panel	Clean, LOW max severity	Clean; pair-level check clear
aroB (primary)	Chromosomal panel	Clean, LOW max severity	Clean pair-level; shares a conserved locus with *C. suis* at forward primer + probe only — no completing reverse-primer hit, so no plausible amplicon

Target / set	Panel	Structure (three-way dimer check)	BLAST specificity
aroB Alt 1	Chromosomal panel	LOW	Clean
aroB Alt 2	Chromosomal panel	LOW	Clean
PSD2 (primary)	Chromosomal panel	LOW	Clean
PSD2 Alt 1	Chromosomal panel	LOW	Clean
group_306	Chromosomal panel	Clean — no hairpin on any of the three oligos	Clean
Plasmid Primary	Plasmid target	0 flags	Clean; matches 7 published *C. trachomatis* lab cloning vectors at 100% identity (expected: these are native-plasmid-derived laboratory reagents, not expected in clinical samples other than as laboratory carry-over); weak partial *C. muridarum* hit on the probe only, no completing pair
Plasmid Reserve	Plasmid target	0 flags	Clean; same lab-vector matches, no non-target *Chlamydia* hits

Cross-interactions between the chromosomal and plasmid oligos of the proposed duplex were also checked (`bin/19` , decision D29). For all 14 possible chromosomal-plus-plasmid pairings (nine cross pairs each), no cross-dimer reaches MEDIUM or HIGH severity on the scale used within each set; the strongest structure Tm is 21.5 °C under default conditions, or 27.0 °C under the project's reaction conditions. This is an oligo–oligo thermodynamic check only: BLAST cross-hits between the targets, detection-channel assignment and real multiplex behaviour remain for laboratory evaluation.

Lessons for computational assay design

- **Recompute inherited metrics rather than propagating reported values.** A Tm, a GC%, a specificity score handed over from a prior pass is a claim, not a fact, until it's regenerated from the actual sequence with the actual tool this project relies on.
- **A single aggregate score can hide the one failure mode it doesn't model.** The 0–100 exclusivity scale rewarded clean BLAST hits and had no term for Tm gap at all — so it could not have caught the error that mattered most, no matter how well it was tuned.
- **An implausible intermediate result is worth stopping for.** A doubled alignment length, a suspiciously round percentage, a candidate that scores identically to another — each is a cheap, early signal that something upstream needs a second look, before it shapes every downstream decision.
- **Check every relevant combination, not just the obvious one.** A three-oligo TaqMan set has three pairwise interactions (F-R, F-Probe, R-Probe); checking only F-R because it's the most familiar leaves two-thirds of the real risk unexamined.
- **Verify quantitative assumptions before allowing them to drive design decisions.** The plasmid's claimed GC% and a proposed Tm target built on it were both wrong, and the fifteen minutes it took to measure the real consensus GC was cheaper than designing against the wrong number.

Scope and limitations

- **Computational scope.** Every result here is a prediction from sequence data and software. Amplification efficiency, analytical sensitivity, specificity in real samples and multiplex performance have not been measured; they are the purpose of the experimental stage.
- **Reference population.** Inclusivity is shown for the 97 primary-tier genomes drawn from 125 RefSeq complete assemblies; lineages absent from RefSeq complete genomes are not represented. The plasmid evidence comes from the 34 plasmid-carrying assemblies in that collection, not from a survey of plasmid diversity.
- **nvCT variant.** The plasmid design is a standard consensus design. The nvCT variant, which carries a deletion in the plasmid, is not claimed to be covered: its published coordinates and accession numbers could not be independently confirmed, so variant-specific design was set aside rather than guessed. The chromosomal locus in the proposed duplex is the intended safeguard against this.
- **Specificity statements.** "Clean" and "no predicted off-target amplicon" describe results under the defined screening conditions (BLAST parameters, databases and the pair-level rule above). They are not a measurement of biological cross-reactivity.

- **The duplex is a proposal.** A second marker is a potential safeguard against false negatives, contingent on inclusivity, primer performance and the behaviour of both targets in one reaction. Oligo–oligo dimer interactions between the two sets were checked; BLAST cross-hits between targets, dye assignment and real multiplex behaviour were not.

Conclusion

The in-silico design phase produced five candidate targets represented by nine primer/probe sets and 27 oligos, all passing the same predefined thermodynamic, structural and specificity criteria. The workflow provides a traceable path from genome QC and target discovery through inclusivity/exclusivity analysis to oligo-level screening, with each design decision documented and reproducible. The panel itself, along with the pipeline scripts that produced it, is committed to the project's GitHub repository. The resulting candidate panel is ready for experimental evaluation, while the underlying workflow can be transferred to target-selection problems in other microbial diagnostic assays.